TERM 2

AGRICULTURE

1) CROP HUSBANDRY
   1.1. PLANT GROWTH
   1.2. VEGETABLE PRODUCTION
2. SOIL FORMATION
3. SOIL CONSTITUENTS

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CROP HUSBANDRY

GERMINATION is:

- The development of a seed into a new plant.
- The process whereby an embryo within a seed begins to develop intoa new plant.

REQUIREMENTS NEEDED FOR SEED GERMINATION

1. *Water*- it is needed for dissolving substances in the seed called enzymes which are needed for growth of the embryo.
   - It also helps to soften the hardcover of seeds so that the young plant can grow easily. When seeds have absorbed water, they swell up and can easily split.
   - It dissolves food kept inside the seed so that the embryo can use it for plant growth.
   - It forms part of new plant cells and tissues.
2. *Correct temperature*- optimum or best possible temperature is needed to have good germination. Each type of plant seed has a certain temperature range for it to germinate.
   - Very hot or cold temperatures will kill the embryo within the seed thus stopping germination to occur.
3. *Air/Oxygen*- for respiration to take place oxygen is needed and this will give the seed energy for germination processes.
   - Energy from food kept in the seed can only be released if oxygen is present.
   - If there is no oxygen, germination cannot take place.
4. *Seed viability*- it is the ability of a seed to germinate and develop into a plant. Seed viability is affected by age of seed, damage by either pests or diseases and the amount of food available in the cotyledons and endosperm.

PLANT GROWTH REQUIREMENTS

1. *Water*- water occupies 80-90% of the plant mass. It is a solvent dissolving plant nutrients.
   - It is also required for making plant food through the process of *photosynthesis.*
   - Water transports dissolved food substances made in the leaves during photosynthesis. The process is called *translocation*.
   - It also helps in cooling the leavesby allowing water vapour and air to move in and out the leaf through stomata.
   - It makes plant cells firm and strong.
2. *Sunlight*- plants use sunlight to make its own food through the process of photosynthesis.

   $$\text{Water + Carbon dioxide} \xrightarrow[by\ chlorophyl]{sunlight} \text{Carbohydrates + Oxygen}$$

   - It helps in the making of chlorophyll (*the green colour in the leaves*) which absorbs light from the sun.
3. *Air-* it is a mixture of gases. Oxygen is used during respiration. Respiration is thebreaking down of stored carbohydrates using oxygen to release Carbon dioxide.

   $$\text{Carbohydrates + Oxygen} \longrightarrow \text{Carbon dioxide + water + energy}$$

   - The carbon dioxide released during respiration is used in photosynthesis or releasedinto the air.

- Oxygen helps to release energy from foods so that plants can use it.

4. *Nutrient or minerals*- although plants are able to make their own food, there is need for plants to get nutrients from the soil. Lack of nutrient element will result in deficiency disease. Farmers should make it a point that plants have correct amounts of the different nutrient elements in the soil.
5. *Suitable (correct) temperature*- not all plants grow well under the same weather conditions, some plants grow well when it is hot while others need cool condition to grow well.
   - Plants cannot grow below a certain temperature. This is called minimum temperature. Plants can also stop growing above a certain temperature and this is maximum temperature.
6. *Absence of harmful things or substances:* salty, oily, rocky or soil that is full of harmful chemicals and pests are not good for plant growth. Farmers must eliminate the harmful things before plant crops to ensure proper plant growth.

VEGETABLE CROPS

IMPORTANCE OF VEGETABLES

- Source of employment
- Source of money (income)
- Source of foreign exchange
- Source of food
- Provide green feed for livestock
- Provide raw- materials for industries
- Provide green manure

Classification of vegetables

- *Leaf vegetables*: spinach, lettuce, celery or chomolia
- *Root vegetables*: carrots, beetroots, radishes,
- *Fruit vegetables*: tomatoes, pumpkins, cucumbers
- *Bulb vegetables*: onions, garlic, leeks
- *Tuber vegetables*: potatoes, sweet potatoes
- *Legume vegetables*: beans, peas

NUTRITIONAL VALUE OF VEGETABLES

Nutritional value means the usefulness of vegetables as food in our bodies. *For eg*:

- Source of vitamins and minerals that protect our bodies from diseases and illnesses. The vitamins and minerals make our bodies heal faster.
- Provide proteins which build and repair worn out tissues and cells. *Eg*, beans and peas.
- They make our food taste better. *Eg*, garlic and onions
- Are source of fibre which make our bowels and stomach work properly. *Eg*, cabbage, carrots, spinach.
- They are rich in energy which is needed to do work.
- Provide oils.

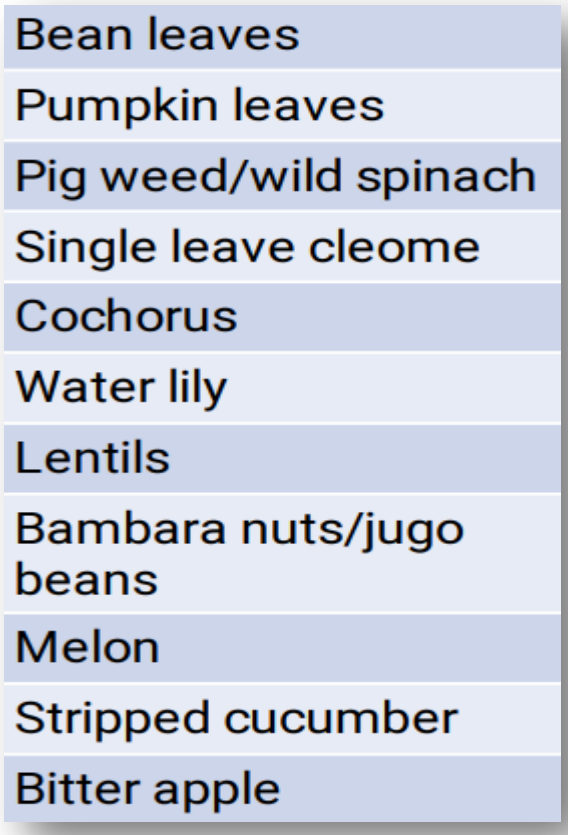
IMPORTANCE OF VEGETABLES TO HIV/AIDS INFECTED PEOPLE

People with HIV/AIDS should eat vegetables because, they:

- Contain proteins that are necessary for growth and repairing worn out tissues and cells.
- Are rich in carbohydrates which give them energy to do work.
- Contain vitamins and minerals which are required by the body to regulate its metabolic functions and to protect against diseases and other illnesses.
- Contain water which is essential for maintaining proper fluid balance, muscle function and heart musclefunction.
- Contains fibre which helps with digestion and excretion.
- Are colourful which encourages people with HIV/AIDS to eat.

INDIGENOUS VEGETABLES

- Those that are native/local to the given country.
- They grow naturally in our country.



EXOTIC VEGETABLES

- Those that are obtained from outside the country.
- Botswana imports them from other countries. Eg:

- Spinach
- Potatoes
- Onions
- Carrots
- Cabbage
- Rape
- Beetroots
- Egg plant
- Cucumber
- Lettuce
- Garlic
- Pepper
- Tomatoes
- Chomolia

ADVANTAGES AND DISADVANTAGES OF INDIGENOUS VEGETABLES

Advantages of indigenous vegetables	*Disadvantages of indigenous vegetables*
- Readily available and cheap as we get them freely in the wild. - Cheaper than exotic fruits. - Money and time is not spent in looking after them. Nature takes care of them. - Source of nutrients needed by the body. - Can grow as rain-fed crops, and they are tolerant to drought - Resistant to most pests - Most are harvested over a long period of time. - Some of them can be harvested more than once.	- Their seeds are not easy to get; therefore, we cannot grow them in our homes. - Not much information is available on how to grow them. - Fresh vegetables are only available during the rainy season. During dry season only dried ones are available. - Their supply is not reliable as some are only found in certain parts of the country only. - Not many types to choose from.

ADVANTAGES AND DISADVANTAGES OF EXOTIC VEGETABLES

Advantages of exotic vegetables	*Disadvantages of exotic vegetables*
- Their seeds are easy to find in shops. - Information on how to grow them is readily available hence easy to grow them. - Available throughout the year unlike indigenous ones. - They are rich in variety of nutrients. - Many different kinds to choose from unlike indigenous ones.	- Some of them are very expensive. - Some of them do not grow well under local weather conditions. (*they are not adapted to local weather conditions*) - They bring with them pest and diseases into our country. - They are very expensive to grow and take care of. - Some of them are not familiar to local people. - Easily attacked by numerous pests and diseases. - Most are harvested only once.

PROBLEMS FACED BY VEGETABLE GROWERS AND THEIR POSSIBLE SOLUTIONS

PROBLEM	SOLUTION
1. Operating costs are high e.g. fuel, salaries forworkers (*commercial farming*)	Forming co-operatives.
2. Bad roads to markets.	Improving local infrastructure.
3. Lack of skilled labour.	LEA and CEDA to intensify training andmentoring of farmers.
4. Stiff or tough competition from neighbouring countries.	Government should limit the importation vegetables and support local farmers. Improve marketing and storage facilities forfarmers.
5. Lack of perennial rivers and reliable rainfall.	Build more to capture water during rainyseason.
6. Outbreak of pests and diseases.	Using chemicals for pests and disease control.

7. Extreme temperatures.	Use of net shading and growing drought resistant crops. Growing vegetables that can do well under semi-arid conditions.
8. Bulkiness and perishability of the products	Refrigerate to increases shelf life
9. Lack of money.	Borrowing money from banks and using government schemes such as CEDA and NAMPAAD.

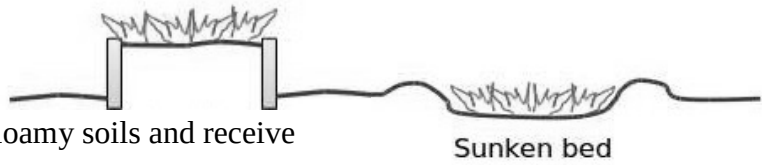
FACTORS TO CONSIDER WHEN SELECTING A GARDEN SITE

1. *Slope of the land*: Flat or gentle sloping land is most suitable or ideal because water will *not* flow away fast or be concentrated in one place but evenly in the plot ensuring proper distribution.
2. *Soil type:* Deep fertile loam soils are often most suitable for crop production than clay or sandy soils because of its characteristics which ensure proper plant growth.
3. *Sources of water supply*: Water is essential for growing crops. Although most crops get water from the rainfall, this may not be enough and a reliable source of water for irrigation is needed e.g. perennial rivers and boreholes. A garden should be where there is always enough water.
4. *Shelter*: Protect crops from effects of strong winds by growing windbreaks around the field or garden.Winds break down crops, increase evaporation and erode soil
5. *Nearness to markets*: Anywhere a buyer and seller meet to exchange goods. Having markets near to where production is taking place help fruits and vegetables not to lose quality or go bad.
6. *Distance from where one lives:* farmers should not stay far from the garden so that it can be easy for them to protect and take care of the crops.
7. *Drainage:* the garden should be in a place which does not hold too little or too much water but enough of it to ensure proper growth.
8. *Presence of light:* the plants should be able to receive enough light for photosynthesis to enhance normal growth.
9. *Size of the garden:* the level of production will determine the size of the garden that the farmer wants to have.

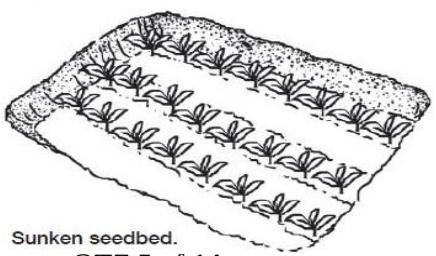
PREPARING PLOT OR BEDS

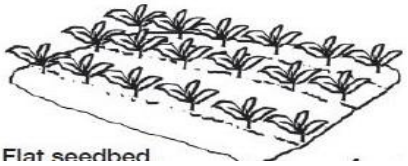
They type of plot a farmer prepare will depend on the type of soil in the garden and the weather conditions in that area. There are 3 types of plots one can make, namely:

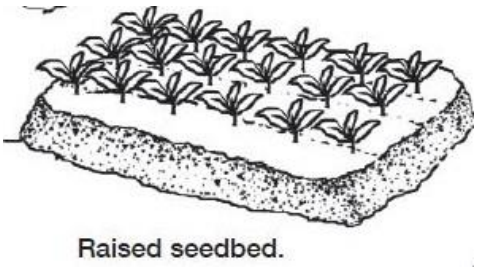
1. Flat bed/plot
2. Sunken bed/plot
3. Raised bed/plot



1. *Flat beds* are suitable in places which have loamy soils and receive moderate rainfall.
2. *Sunken plots* used in places which receive little or rainfall so that it can catch water when it rains.
3. Raised plots are used in places which have clays soil and receive a lot of rainfall. This will ensure that water does not stand on the plot for a long times which can cause plants to rot and also removes air from the soil.







PREPARING A PLOT

After deciding which type of plot to make, you can start doing the following:



- Clear the plot to remove pest and weeds. Tools such as spade and hoe can be used.



- Dig the soil to make it loose for water, air and plat roots to move or penetrate easily. Tool such as spade, digging fork and pick can be used to dig the soil.



- Add manure or fertilizer to the beds to make the soil fertile. Then mix it well with soil.



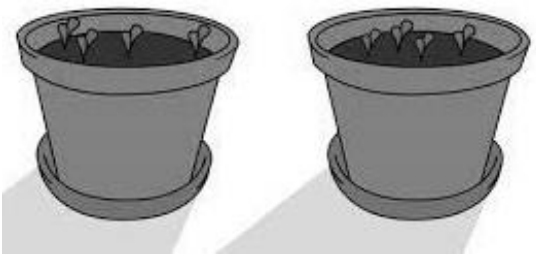
- Level the soil with rake. This is to make sure that water is well distributed in the plot. If the plot is not leveled water will flow to one side and other plants will not receive water.

METHODS OF PLANTING





- *Direct sowing*- seeds are sown in to main beds where they will grow until they are harvested.
  - Vegetables such as carrots and beetroots are usually sown directly.
  - Planting seeds directly into the soil usually uses drilling and broadcasting methods.



- *Indirect sowing*- planting seeds first in nurseries and later transplanted to permanent plots where they will grow until they are harvested.
  - Vegetables such as tomatoes and cabbage are fist sown in nurseries.
  - The seedlings are given special care until they are strong enough to withstand the weather conditions outside the nursery is called *transplanting*.

SPACING OF PLANTS /SEED RATE

Plants need to be spaced correctly so that they can grow well. If they are crowded, they will compete for water, air, nutrients and sunlight and if they are scattered the harvest will be low. The amount of spacing is determined by:

1. Soil fertility
   - If the soil has little or no nutrients, the spacing between plants must be made bigger to reduce competition.
   - If the soil is fertile, plants can be spaced closer to each other because there is abundance of nutrients.
2. Availability of water
   - If there is little water, plants should be spaced further from each other to reduce competition.
   - If water is plentiful, plants can be planted closer to each other.
3. How big the crop will grow
   - Plants like carrots can be placed closer to each other because they do not grow bigger as compared to cabbage which must be placed further apart because they grow bigger.
4. Size of the implement used in the farm
   - If implements or machines are to be used in the farm for management practices, there should be enough space between the plants to enable implements or machines to work without destroying the crops.
5. Reasons for growing the plant
   - Beans grown to provide nitrogen fixing bacteria are spaced much closer to each other than those grown for subsistence or commercial use.

The following table shows some suggestion of how much space to leave between plants

VEGETABLE	SPACING WITHIN ROWS	SPACING BETWEEN ROWS
Beetroot	7-10 cm	30-50 cm
Cabbage	35-50 cm	70-100 cm
Carrots	5 cm	30-50 cm
Onions	7 cm	10-30 cm
Lettuce	20 cm	30-60 cm
Peas	20 cm	30-60 cm
Spinach	20 cm	36-60 cm
Tomatoes	40-50 cm	100-180 cm
Green pepper	40-60 cm	90-100 cm
Runner beans	10 cm	90-120 cm

DEPTH OF SOWING

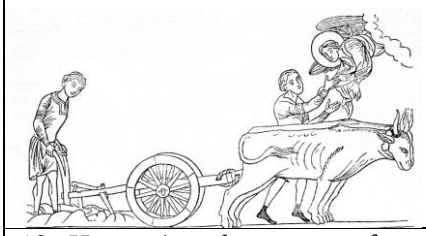
- Depth of planting seeds depends on its size and the type of soil in which it is planted.
- Seeds are not of the same size, *ie*: beans are bigger hence they are planted a bit deeper whereas carrots seeds are small therefore, they are planted a shallow.
- If small seeds are planted too deep they will not emerge because they will use all energy before reaching the soil surface.

The depth of sowing is determined by calculating 3-5 times the diameter of eachgiven seed. This means that if a seeds is 1mm in diameter, it should be planted between 3-5 mm.

VEGETABLE MANAGEMENT PRACTICES

These are activities done to ensure the proper growth of vegetables.

MANAGEMENT PRACTICE	REASONS FOR DOING THEM
 1. *Watering* – it is done using a watering can.	➢ Cools the plant ➢ Transport nutrients around the plant ➢ It forms part of new plant cells and tissues. ➢ Vital for photosynthesis.
2. *Mulching*- covering the bed with dried material such as grass or twigs. 	➢ Regulates soil temperature *(keeps the soil cool when it's hot and warm when it's cold)* ➢ Reduces water loss through evaporation. ➢ Prevents weed growth because weeds cannot grow without sunlight. ➢ Prevents soil erosion. ➢ Mulch rots and forms humus. ➢ Prevents pests from eating seeds and emerging seedlings. *Mulch must be removed as soon seedlings emerge so that they can be able to receive sunlight.*
 3. *Transplanting* - moving seedlings from a nursery to a permanent plot.	➢ This done to give other seedlings special care before being transplanted to a permanent plot. The special is given in a nursery. After that they are transplanted when they are strong enough. ➢ *Hand trowel* is used to transplant.
4. *Shading*- providing plants with shade. 	➢ Shading protects plants from being killed by heat from the sun. ➢ The shaded soil stays cool and loses little water through evaporation.
 5. *Thinning*- removing extra and weak seedlings so that those that remain in the bed are not crowded.	➢ So that remaining seedlings can have enough space. ➢ To reduce completion between the growing seedlings. ➢ To enhance yield as the remaining crops will grow well. ➢ It can be done using *hands.*
6. *Prunning*- Getting rid of dead and unproductive plant parts which are useful or productive. 	➢ So that healthy and productive plant parts remain. ➢ Enhance growth of the plant. ➢ So that food, water and air are not wasted on dead and unproductive plant parts. ➢ Can be done to certain plants which are used to decorate. ➢ A very *sharp knife, a scissor* can be used.

 7. *Pest and disease control*- chemicals such pesticides (*for killing pests*) and weedicides/herbicides (*for killing weeds*) can be used.	➢ If not controlled plants will grow poorly and eventually die. ➢ Pest and diseases cause poor yield. ➢ We get poor crop quality from plants attacked by pests and diseases. ➢ A *knap sack sprayer* is used to apply the chemicals.
8. *Weeding*- The removal of unwanted plants in the plot. The unwanted plants are called . 	➢ Weeds compete with crops for space, water, food, air and sunlight making crops grow poorly. ➢ Weeds spread disease and pest to plant which might end up killing them. ➢ They reduce yield as the plants will grow poorly.
9. *Cultivation* - tilling or soil refinement, is the act of digging into or cutting up an existing soil bed to better prepare it for planting 	➢ It breaks down hard soil surface to make soil loose so that roots, air and water can penetrate the soil easily. ➢ Weeds are killed during the process. ➢ Underground pest are exposed and killed during the process.
10. *Harvesting*- the process of collecting ripe crops. (Harvesting is the operation of gathering the useful/edible part or parts of the plant). 	➢ For sale and consumption. ➢ Crops are harvested when they are mature or fully grown. ➢ If they are not harvested on time they will lose quality or get spoilt.
11. *Gapping*- Filling in gaps left by plants which either do not germinate or dried out	➢ To increase yield. ➢ This can reduce spoilage ➢ improve quality and ensure that you have the best quality produce for your customers

METHODS OF PRESERVING VEGETABLE

- Vegetables are very perishable; therefore, they need to be preserved.
  - Preserving vegetables means protecting them getting damaged or spoilt easily.
- If vegetables are preserved: - they will last longer.
  -they will be available throughout the year
  -they will not lose nutrients.

Vegetables can be preserved in the following ways:

A. Dehydration

- Removing moisture from vegetable or drying them.
- Vegetables are left in the sun after harvesting so that moisture can be removed. A lot of vegetables can be dried such onions, spinach, bean leaves, pig weed, cochorus etc.
- When moisture has been removed from vegetables, micro-organisms such as bacteria will not grow. Micro-organisms cause vegetables to rot.

B. Refrigeration

- Keeping vegetables in a refrigerator (fridge) where temperatures are very low to preserve them.
  A cold environment slows the growth of micro-organisms hence vegetables can be stored a longer periods of time. A lot of vegetables can be refrigerated such as*: - Plums - Peaches - Apple - Grape - Mango - Apricot - Raisins*

C. Freezing
- Putting vegetables in freezer where temperatures are extremely cold.
- There will be no micro-organisms and vegetables can be stored for a very long, long time. Most vegetables can be refrigerated such: *strawberry*

D. Canning
- This involves cleaning, slicing and putting them in cans.
- Fruits that can be canned include: *Pears - Pineapple - Guava - Peaches*

VEGETABLE FARM RECORDS
- If the information is about what was bought or sold, costs and profit or loss, then it is called a *financial record*.
- If the information is about important things or events that have taken place in the garden when growing vegetables, then it is called a *diary record*.
- If the information kept is showing the quantity of the vegetable crop harvested, then it is called production record.

1) FINANCIAL RECORD

It shows income (money coming in), expenditure (money used to buy items) total cost of the items bought in the farm.

EXPENDITURE			*INCOME*		
Date	Item	Cost	Date	Item	Cost
10/04/21	Spinach seeds	P100.00	01/06/21	Spinach sold	P400.00
10/04/21	Tomato seeds	P80.00	01/06/21	Sold tomatoes	P600.00
12/04/21	Kraal manure	P50.00	02/06/21	Sold spinach	P450.00
13/04/21	2 bag fertliser	P200.00	03/07/21	Sold tomatoes	P400.00
14/06/21	Pesticides	P60.00			
	Labour	P300.00			

2) PRODUCTION RECORDS

It shows what was produced, how much were produced, date of harvest and some comments concerning production.

Crop harvested	*Date harvested*	*yield*	*Comments*
Beetroots	11/02/22	100 kg	Some beetroots were too small
Spinach	15/02/22	200 kg	Insects ate some leaves.

3) DIARY RECORD

Year: 2022
Vegetable crop planted: Runner beans
Date of planting: 11/02/22
Depth of planting: 5 mm
Method of planting: direct sowing
Date of seedling emergence: 18/02/22
Spacing within rows: 10 cm
Spacing between rows: 100 cm

Date	***Activities done***	***Comments***
11/08/21	A plot 4 m x 1 m was dug using a spade then leveled using a rake.	The soil was hard to dig. Took an hour to dig properly.
12/08/21	Manure was mixed with the soil.	The manure had a lot of dirt.
12/08/21	Watering was done using a watering can.	
13/08/21	Runner bean seeds were planted	Seeds were planted 2 per hole.

REASONS FOR KEEPING RECORDS

- They help the farmer to remember which activities were carried out
- They help the farmer to know when an activity was carried out and when to do it again
- They help the farmer to compare production from one year to the next
- They help the farmer to identify problems and come with possible improvements in production
- They help the farmer to manage his or her finances, by determining if he is making a profit or loss.
- They help farmers to obtain loans from financial institutions
- They help farm advisors or extension officers to give better advice

MARKETING OF VEGETABLES

Marketing refers to all those processes involved in the preparation of raw agricultural product as produced at the farm, to consumer goods.

Marketing of fruits involves:

- Advertising: Making products known to consumers.
- Harvesting: collecting mature and ripe crop vegetables.
- Cleaning: removing dirt from vegetables by washing them with clean water.
- Sorting: grading vegetables according to size.
- Packaging: putting vegetables in bundles, plastics and boxes.
- Pricing: putting prices in vegetables.
- Labeling: tagging vegetables.
- Storage: putting shelves and storeroom.
- Preservation: protecting them getting damaged or spoilt easily
- Processing them to make others products such as fruit juice
- Selling

Market places for fruits include:

- Individuals - Wholesalers - Retailers - Cooperative societies
- Traders - Government departments (schools, hospitals, etc)

SOIL FORMATION

Soil is formed through the process called *weathering.*

Weathering- this is the breaking down of rocks into smaller pieces to form soil.

Types of weathering

1) Physical weathering- the breaking down of rocks into smaller pieces without changing the chemical composition of the rock. *No change in chemical composition means that there is no change in colour of the original rock.*

Agents of physical weathering; *temperature, water and wind.*

a. *Temperature*- During the day the temperatures are high causing therock to heat up and expand. When temperatures drop at night, the rocks cool and crack. This continuous heating and cooling overtime, causes the rock to break intosmaller pieces.
b. *Water*- flowing water across the land carries with itself materials like stones and sand with it. These materials scrape and knock against any hard rocks that the water flows over causing small pieces to break off and overtime soil particles are formed. This also happens when the wind is blowing.

c. *Wind* - As wind blows it carries pieces of rocks which break down into small particles by knocking against each other.

2) Chemical weathering- this is the breaking down of rocks into smaller pieces with a change in chemical composition of the rock. *Change in chemical composition means that there is change in colour of the original rock.*

Agents of chemical weathering*; air [oxygen and carbon dioxide] and water.*

a. O*xidation*- this when minerals combines with oxygen to form an oxide. Minerals in therock combine with oxygen to form oxides. Red-coloured rocks contain iron oxideswhich have been formed by the process of oxidation. When minerals in rock oxidize, they become less resistant to weathering.
b. *Carbonation*- occurs in limestone. Carbonation is formed when rainwater dissolves the carbon dioxide in air to form a weak acid called carbonic acid.

Rainwater + carbon dioxide = Carbonic acid.

When rain, containing carbonic acid falls on the rock with a lot of limestone, it willreact with limestone to form calcium carbonate.

Carbonic acid + limestone = Calcium carbonate

All the reaction above causes the rock to be soft and breaks easily.

3) Biological weathering- this is the breaking down of rocks through the activities of living organisms.

Agents of Biological weathering*; plants, animals and human activities.*

a. *Plants*- trees with their roots grow in between the cracks of rocks, causing the rock towiden up and break down the rocks.
b. *Animals*- animals which burrow through soil like earthworms, termites, mole rat andyellow mongoose can help form soil.
c. Human activities like mining, cultivation, construction of buildings, railways androads reduce the size of rocks by breaking them up to smaller pieces.

SOIL

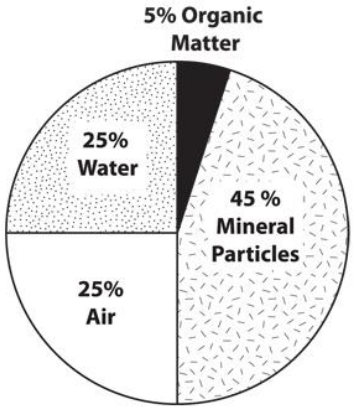
- Soil is top layer of the earth in which the plants grow, derive their nutrients, water and get anchorage
- Soil is a medium on which plants grow.

The importance of soil

- It gives plants nutrients, water and air needed for growth
- It holds plants up straight as they grow up
- It is a habitat for some animals
- It is used to build houses-
- It is where we grow crops and produce fruits for eating.
- It provides grazing land and grasses for animals
- It has resources such as wood, minerals and grass.

SOIL CONSTITUENTS

- These are substances that make up soil, eg:
    - Inorganic matter or mineral matter (forms large part of the soil)
    - Organic matter – remains (decomposed parts) of plants and animals
    - Water
    - Air



1. **Rock particles or mineral matter**
   This is a non-living matter that makes up 45% of the soil volume. The size of the soil particles varies from a diameter of 2mm in gravel to less than 0.002mm in clay soil.

2. **Organic matter**
   About 5% of the soil volume consists of organic matter made up of decomposing plants and animal and also living organisms. Soils with dark brown colour indicate that it contains a lot of **humus**.

3. This occupies the space between the soil particles. It is made up of 25% soil volume. Air mainly consists of nitrogen, oxygen and carbon dioxide as well as other small gases. Soils with large particles have more air spaces compared to soils with smaller particles.

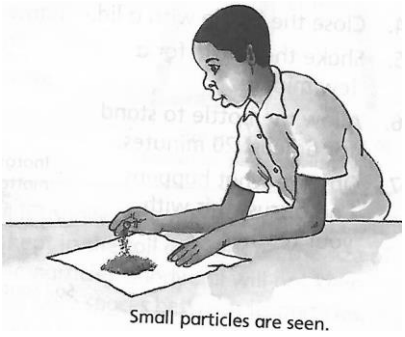
4. **Water**
   Water occupies the space between rock particles making up 25% of soil volume. Afterheavy rains there will be more water in the soil, while after a dry period there will be much less.

INVESTIGATING SOIL CONSTITUENTS

Experiment 1

- **Finding out if the soil has inorganic matter**
  - Collect dry sample of soil
  - Spread it on a sheet of paper and study it carefully
  - *What do you see?*
  - ✓ You will the see the particles of broken down rocks.

⇒ This is the solid part of soil called inorganic matter


Small particles are seen.

Experiment 2



Soil contains organic and inorganic matter

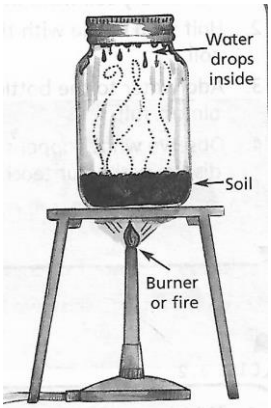
- **Finding out if the soil has organic matter**
  - Collect a dry sample of soil
  - Put the dry sample in a bottle with a lid
  - Fill up the bottle with water and close it
  - Shake the bottle for a few minutes
  - Allow the bottle to stand for at least 15 minutes.
  - ✓ You will see some materials will float at the top of the water

⇒ These materials are referred to as organic matter

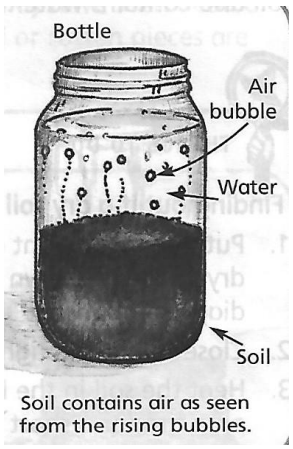
Experiment 3

- **Finding out if soil is made up of water**
  - Put a small amount of dry soil in a dry bottle and close the bottle tightly with a lid
  - Heat the soil in the bottle slowly on a fire.
  - Observe the bottle after at least 20 minutes
  - ✓ You will see water droplets inside the bottle

⇒ The droplets of water shows the presence of water in the soil



Dry soil contains water as shown by the water droplets.



Soil contains air as seen from the rising bubbles.

Experiment 4

- **Finding out if soil has air**
  - Collect dry soil in the bottle
  - Half fill the bottle with water
  - ✓ You will see air bubbles rising from the soil

⇒ This shows the presence of air in the soil